KRISHNA TANWAR

Software Developer | Python & Backend Engineering | Relational Databases & SQL | AI/ML

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EDUCATION

Bachelor of Technology (B.Tech) — Computer Science & Engineering

2023 – 2027 (Expected)

GLA University · Specialization in Artificial Intelligence & Machine Learning (AI/ML)

- Core Coursework: Data Structures & Algorithms, Object-Oriented Programming (Java/Python), Database Management Systems (SQL), Applied Machine Learning, Operating Systems & Computer Networks.

PROFESSIONAL EXPERIENCE

Graduate Engineer Trainee — SMAAC Net Solutions

Present

Full-time Software Development & Enterprise Engineering

- Contributing to backend software development, business logic modules, and database management workflows.
- Adhering to clean code architecture, code review standards, and agile sprint delivery cycles.
- Working with relational schemas and optimizing query operations for enterprise systems.

Data Analyst Intern — AtliQ Technologies

Nov 2025 – Feb 2026

Business Intelligence & Transactional Analytics

- Engineered executive Power BI dashboards with DAX measures and Power Query ETL pipelines across 100K+ transactional records.
- Designed robust star-schema relational models connecting sales, customer demographics, and financial datasets.
- Formulated SQL queries for cohort retention, margin analysis, and sales forecasting, enhancing executive reporting accuracy.

SELECTED TECHNICAL PROJECTS

RAG AI Teaching Assistant

`Python` `Ollama` `Whisper` `Vector Embeddings` GitHub Repo

- Engineered an automated video tutorial RAG assistant using Whisper speech-to-text and local Ollama vector embeddings.
- Chunked subtitle transcripts with semantic windowing, returning precise video timestamp citations for student questions.

Smart Traffic Violation AI Platform

`Python` `YOLO` `EasyOCR` `React` `SQLite` GitHub Repo

- Developed full-stack violation detection platform with YOLO vehicle tracking, license plate OCR, and automatic rule-checking.
- Integrated SQLite case logging database and React dashboard for audit history and automated citation generation.

Real-Time Video Multi-Class Detection System

`Python` `OpenCV` `YOLO` `Flask` GitHub Repo

- Implemented multi-threaded real-time streaming pipeline supporting vehicle detection, human tracking, and motion detection.
- Built configurable performance profiles (fast, balanced, accurate) with web-based feed monitoring.

Business Insights 360 Enterprise BI

`Power BI` `DAX` `SQL` `Data Modeling` GitHub Repo

- Created unified cross-functional BI suite for AtliQ Hardwares spanning Finance, Sales, Marketing, and Supply Chain.
- Engineered dynamic time-intelligence DAX measures replacing manual spreadsheets with real-time analytics.

TECHNICAL SKILLS

Languages:	Python, SQL, Java, JavaScript, HTML5, CSS3
Backend & Frameworks:	Flask, REST APIs, Object-Oriented Architecture, System Design Fundamentals
Databases & Storage:	MySQL, PostgreSQL, SQLite, Database Normalization, Schema Design, Query Optimization
AI / ML & Computer Vision:	RAG (Retrieval-Augmented Generation), Whisper, Ollama, OpenCV, YOLO, TensorFlow, Vector Embeddings
Analytics & Tools:	Power BI, DAX, Power Query, Pandas, NumPy, Excel, Git, GitHub, VS Code